

Spider-Inspired Multi-Legged Robot for Stable Climbing on Uneven Slopes in MuJoCo Simulation

Hoyoung Cho

Abstract

Locomotion on uneven and inclined terrain remains a difficult problem in robotics because rough surfaces reduce traction, disturb posture, and increase slipping. This project investigates whether a spider-inspired multi-legged robot can improve stability on uneven slopes in simulation. An eight-legged MuJoCo robot was developed with a low body posture, distributed leg support, and a slow alternating gait. The simulation environment consisted of a flat base plane, a 20° slope, a flat platform at the top of the slope, and uneven box obstacles distributed across the incline. Performance was evaluated using body x-position, body pitch, body height, and cumulative slip count over time. Over 68.03 s of simulation time, the robot advanced from -1.80 m to 7.38 m in x-position, climbed from 0.43 m to 2.77 m in body height, reached a minimum pitch of -25.49° , and accumulated 10 slip events. These results show that the spider-inspired robot was able to climb the uneven slope and maintain forward progress, but slipping and large body pitch changes remained significant limitations.

1 Introduction

Locomotion on rough and inclined terrain remains a major challenge in robotics. Traditional wheeled robots perform well on smooth ground, but their performance often decreases when the surface becomes uneven, sloped, or discontinuous. Legged robots are more suitable for these environments because they can adapt foot placement, maintain posture, and distribute support across multiple contact points. Because of this, legged locomotion is often used when terrain is difficult or unpredictable.

This project focuses on a spider-inspired robot for climbing uneven slopes in simulation. Spiders are a useful biological model because they remain stable by distributing support across many legs rather than relying on only a few ground contacts. Instead of reproducing exact spider anatomy, this work applies the spider’s distributed-support principle to a simplified robot model and tests whether that idea improves locomotion on inclined terrain.

The objective of this project is to determine whether a spider-inspired multi-legged robot can move stably on an uneven slope in MuJoCo simulation. The final study evaluates that idea using an implemented robot model, recorded simulation data, and generated figures showing forward progress, body pitch, body height, and slip count.

2 Biological Inspiration

The main biological principle used in this project is the way spiders maintain stability by supporting the body with several legs at the same time. When one leg is lifted and moved, the remaining legs continue to support the body, which helps preserve balance during locomotion. In this project, that idea is translated into an eight-legged robot with a wide stance, a relatively low body, and a gait designed to keep multiple feet in contact with the ground throughout the climb.

The goal of the design was not to build an anatomically exact spider. Instead, the project borrowed the spider’s stability strategy and implemented it in a simplified engineering model that could be tested in simulation.

3 Robot Design

The final robot was implemented in MuJoCo as an eight-legged model with a central rectangular body and two joints per leg. The body was modeled as a box with MuJoCo size parameters (0.38, 0.17, 0.06) m and mass 0.65 kg. The body friction was set to (1.0, 0.3, 0.3). The spider body was initialized at position (−1.9, 0, 0.98) in the XML model.

Each of the eight legs had a yaw joint and a pitch joint. The yaw joint range was $\pm 70^\circ$, and the pitch joint range was $\pm 55^\circ$. Each coxa segment was modeled as a capsule with size 0.020 and mass 0.025 kg, while each lower leg segment used size 0.018 and mass 0.035 kg. Each foot was modeled as a sphere with radius 0.030 m, mass 0.015 kg, and friction (8.0, 2.5, 2.5). The actuators were implemented as position actuators with $k_p = 95$. The yaw actuator control range was ± 1.22 rad, and the pitch actuator control range was ± 0.96 rad.

The terrain consisted of a large base plane with friction (2.0, 0.5, 0.5), a 20° uphill slope defined by a box with size (4.8, 1.55, 0.08), and a flat top platform with size (1.25, 1.55, 0.08). Both the slope and the top platform used friction (4.0, 1.2, 1.2). To create uneven terrain, 35 obstacle boxes and 22 edge obstacle boxes were placed on the slope. The obstacles used friction (5.0, 1.5, 1.5).

The overall design emphasized stability rather than speed. The body was kept relatively low, the stance was wide, and the multiple legs were intended to distribute support across the terrain while the robot climbed uphill.

4 Simulation Setup and Methodology

The project was completed entirely in MuJoCo simulation. The model used gravity $g = 9.81$ m/s², timestep 0.002 s, and 100 solver iterations. The default geom friction was (3.0, 1.0, 1.0), and joint damping was set to 1.5.

The simulation script used the explicit control parameters listed in Table 1. The robot mass used in the force model was 0.65 kg, the slope angle was 20.0° , and the effective terrain friction coefficient used in the force calculation was $\mu = 4.0$. A loss factor of 0.20 was included in the uphill force term to account for inefficiency during climbing. The target uphill speed was set to 0.22 m/s.

Table 1: Main simulation and control parameters

Parameter	Value
Gravity	9.81 m/s ²
Timestep	0.002 s
Solver iterations	100
Robot mass	0.65 kg
Slope angle	20.0°
Effective friction coefficient μ	4.0
Loss factor	0.20
Target uphill speed	0.22 m/s
Speed gain K_P	12.0
Side damping gain $K_{D,\text{side}}$	16.0
Vertical damping gain $K_{D,\text{vertical}}$	2.0
Maximum uphill force	4.5 N
Maximum side force	2.0 N
Maximum vertical force	1.0 N
Stop position	$x = 7.40$ m
Brake gains (x, y, z)	(10.0, 10.0, 2.0)
Maximum brake force	5.0 N
Gait frequency	0.35 Hz
Yaw amplitude	0.26 rad
Lift amplitude	0.32 rad
Logging interval	0.05 s
Slip speed threshold	0.60 m/s
Slip cooldown time	0.80 s

From these values, the main slope-related force terms were

$$F_n = mg \cos \theta = 5.99 \text{ N},$$

$$F_{g,x} = mg \sin \theta = 2.18 \text{ N},$$

and

$$F_{f,\text{max}} = \mu F_n = 23.97 \text{ N}.$$

The robot used an alternating leg grouping. Group A consisted of L1, R2, L3, and R4, while Group B consisted of R1, L2, R3, and L4. The gait was designed to maintain frequent ground contact while still producing forward motion up the incline. Simulation data were logged every 0.05 s, and slip was counted when the horizontal speed of a contacting foot exceeded 0.60 m/s with a cooldown of 0.80 s.

5 Performance Metrics

Four metrics were used to evaluate performance.

The first metric was **body x-position over time**, which measured net forward movement and verified that the robot generated real locomotion instead of moving its legs in place.

The second metric was **body pitch over time**, which measured how much the robot body rotated while interacting with the slope and obstacles.

The third metric was **body height over time**, which measured vertical climbing progress and confirmed that the robot moved to higher terrain.

The fourth metric was **cumulative slip count over time**, which measured how often the robot lost traction while moving uphill.

6 Results

The simulation ran for 68.03 s and produced measurable forward motion, body pitch changes, body height changes, and repeated slip events.

As shown in Fig. 1, the robot’s body x-position increased from -1.80 m at the start of the run to 7.38 m at the end. This corresponds to a net forward displacement of approximately 9.18 m. The x-position curve increased steadily through most of the simulation before leveling off near the end, which indicates that the robot produced sustained forward locomotion and eventually reached the stopping condition near the top of the terrain.

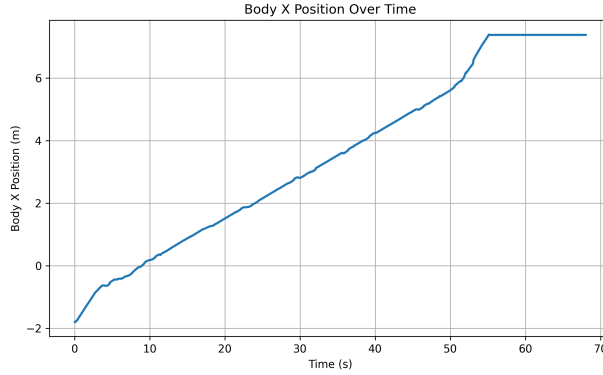


Figure 1: Body x-position over time. The robot moved from -1.80 m to 7.38 m, showing sustained forward progress through the terrain.

The body pitch results are shown in Fig. 2. At the beginning of the simulation, the pitch remained close to 0° , which is consistent with motion on flat terrain. During the slope traversal, the pitch shifted strongly negative and reached a minimum value of -25.49° . The maximum observed pitch was 0.58° . The oscillations and spikes in the pitch curve indicate that the robot experienced significant orientation changes while interacting with the incline and uneven obstacles.

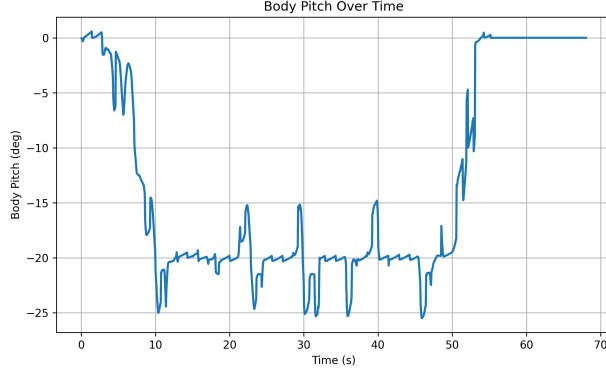


Figure 2: Body pitch over time. The pitch remained near 0° on flat terrain and then shifted as low as -25.49° during the slope traversal.

The body height results are shown in Fig. 3. The body height increased from 0.43 m at the beginning of the run to 2.77 m at the end. This is an increase of approximately 2.34 m, confirming that the robot climbed from lower terrain to a higher region. The height curve rose steadily through the climbing phase and then stabilized after the robot reached the upper area.

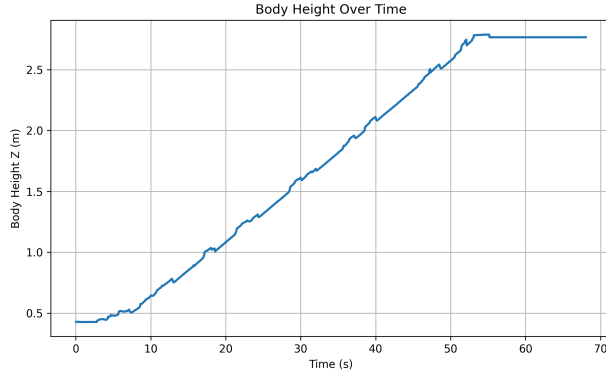


Figure 3: Body height over time. The body height increased from 0.43 m to 2.77 m, confirming successful climbing to higher terrain.

The cumulative slip count is shown in Fig. 4. The slip count increased from 0 to 10 over the course of the run. This shows that although the robot was able to continue moving uphill, it repeatedly lost traction while traversing the uneven slope.

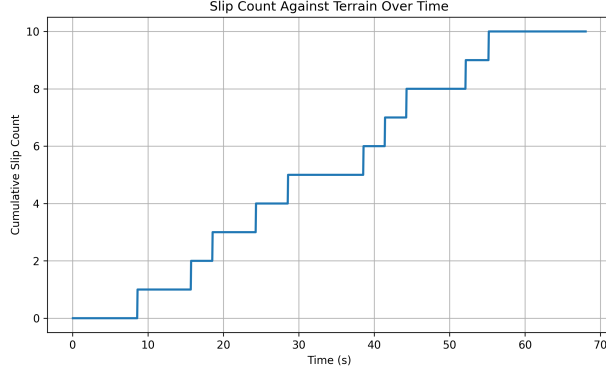


Figure 4: Cumulative slip count over time. The robot accumulated 10 slip events during the run, indicating repeated traction loss on the uneven slope.

Overall, the simulation results show partial success. The robot maintained forward progress and climbed the terrain successfully, but it still experienced repeated slipping and large body pitch changes during the climb.

7 Discussion

The results show that the spider-inspired design was successful in producing uneven-slope locomotion in simulation. The forward displacement of 9.18 m and the height increase of 2.34 m both indicate that the robot was capable of climbing the terrain rather than simply oscillating in place. These are meaningful outcomes because they show that the distributed-support concept can be translated into a functional simulated robot.

However, the results also reveal clear weaknesses. The body pitch reaching -25.49° indicates that the robot posture was strongly disturbed by the slope and obstacle interaction. Similarly, the final slip count of 10 shows that traction was not reliably maintained throughout the run. The robot could still progress uphill, but it did so while repeatedly losing grip.

These findings suggest that the spider-inspired morphology helped provide enough support for the robot to keep moving, but morphology alone was not enough to guarantee stable climbing. The results indicate that gait tuning, better contact timing, and improved foot-terrain interaction would likely reduce both pitch disturbance and slip count. In that sense, the project demonstrates that spider-inspired distributed support is a promising design principle, but also shows that successful slope locomotion depends on both physical structure and control quality.

8 Limitations

This project has several limitations. First, it was performed entirely in simulation, so the contact, friction, and slip behavior may not exactly match what would happen in a physical robot. Second, the gait was not exhaustively optimized, which likely contributed to the large pitch oscillations and repeated slip events. Third, the robot was biologically inspired rather than anatomically exact, since the goal was to test the general stability principle of spider locomotion rather than build a complete spider replica.

Because of these limitations, the final model should be interpreted as a proof of concept rather than a complete terrain locomotion solution.

9 Conclusion

This project presented a spider-inspired eight-legged robot for climbing uneven slopes in MuJoCo simulation. The model used a 0.65 kg rectangular body, a 20° slope, multi-legged distributed support, and a slow alternating gait. The final results showed that the robot advanced 9.18 m in the x direction, increased its body height by 2.34 m, and successfully climbed the terrain. At the same time, the robot reached a minimum body pitch of -25.49° and accumulated 10 slip events, showing that stability and traction were not fully controlled.

Overall, the project demonstrates that a spider-inspired multi-legged robot can traverse an uneven slope in simulation, but it also highlights the need for improved gait tuning and contact control in future work.

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